

3.6 Further Mechanics and Thermal Physics

1. Circular Motion

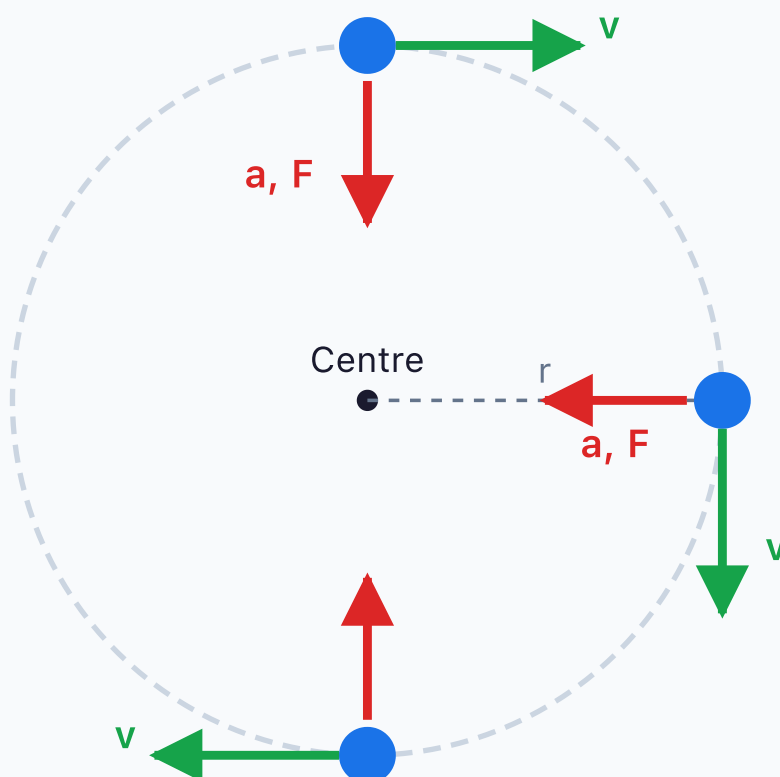
Uniform circular motion: Motion in a circle at constant speed. Although the speed is constant, the velocity is continuously changing (because the direction changes), so the object is accelerating. This acceleration is always directed towards the centre of the circle.

Key Quantities

Quantity	Symbol	Equation	Units
Angular displacement	θ	$\theta = s/r$ (arc length / radius)	rad
Angular velocity	ω	$\omega = 2\pi f = 2\pi/T = v/r = \Delta\theta/\Delta t$	rad s^{-1}
Linear (tangential) speed	v	$v = \omega r = 2\pi r/T$	m s^{-1}
Centripetal acceleration	a	$a = v^2/r = \omega^2 r$	m s^{-2}
Centripetal force	F	$F = mv^2/r = m\omega^2 r$	N
Period	T	$T = 2\pi r/v = 2\pi/\omega = 1/f$	s
Frequency	f	$f = 1/T = \omega/(2\pi)$	Hz

One complete revolution = 2π radians = 360 degrees. To convert: angle in radians = angle in degrees $\times \pi/180$.

Circular Motion -- Vectors



v = velocity (tangent) | a, F = centripetal acceleration/force (towards centre)

The centripetal acceleration (and force) is always directed towards the centre of the circle. It is perpendicular to the velocity at all times. Because the force is always perpendicular to the velocity, **no work is done** by the centripetal force, and the kinetic energy (and speed) of the object remains constant.

Exam Tip: Centripetal force is NOT a separate type of force. It is the *resultant* of real forces (gravity, tension, friction, normal contact) that happen to point towards the centre. Never label "centripetal force" as a separate force on a free body diagram. Instead, identify which real force (or combination of forces) provides the centripetal force.

Common Examples of Circular Motion

Scenario	What provides the centripetal force
Car going around a flat bend	Friction between tyres and road
Car on a banked track	Horizontal component of the normal reaction force (plus friction if needed)
Satellite orbiting Earth	Gravitational attraction
Ball on a string (horizontal circle)	Horizontal component of tension in the string
Electron orbiting a nucleus (classical model)	Electrostatic (Coulomb) attraction

Worked Example: A 0.20 kg ball is attached to a string and moves in a horizontal circle of radius 0.80 m at a speed of 3.0 m s^{-1} . Calculate (a) the centripetal acceleration, (b) the centripetal force, (c) the angular velocity.

Solution:

(a) $a = v^2/r = (3.0)^2/0.80 = 9.0/0.80 = \mathbf{11.25 \text{ m s}^{-2}}$ (11 m s^{-2} to 2 s.f.)

(b) $F = ma = 0.20 \times 11.25 = \mathbf{2.3 \text{ N}}$

(c) $\omega = v/r = 3.0/0.80 = \mathbf{3.75 \text{ rad s}^{-1}}$ (3.8 rad s^{-1} to 2 s.f.)

Check: $F = m\omega^2 r = 0.20 \times (3.75)^2 \times 0.80 = 0.20 \times 14.06 \times 0.80 = 2.25 \text{ N}$. Consistent.

Vertical Circular Motion

For an object on a string moving in a vertical circle, the speed and tension vary around the loop:

- **At the top:** $mg + T = mv^2/r$ (weight and tension both point toward centre). The minimum speed occurs when $T = 0$: $v_{\min} = \sqrt{gr}$.
- **At the bottom:** $T - mg = mv^2/r$ (tension points toward centre, weight points away). Tension is greatest here.

2. Simple Harmonic Motion (SHM)

Simple harmonic motion (SHM): An oscillation in which the acceleration is:

- (1) always directed towards a fixed equilibrium position, and
- (2) directly proportional to the displacement from that equilibrium position.

These two conditions are expressed by the defining equation:

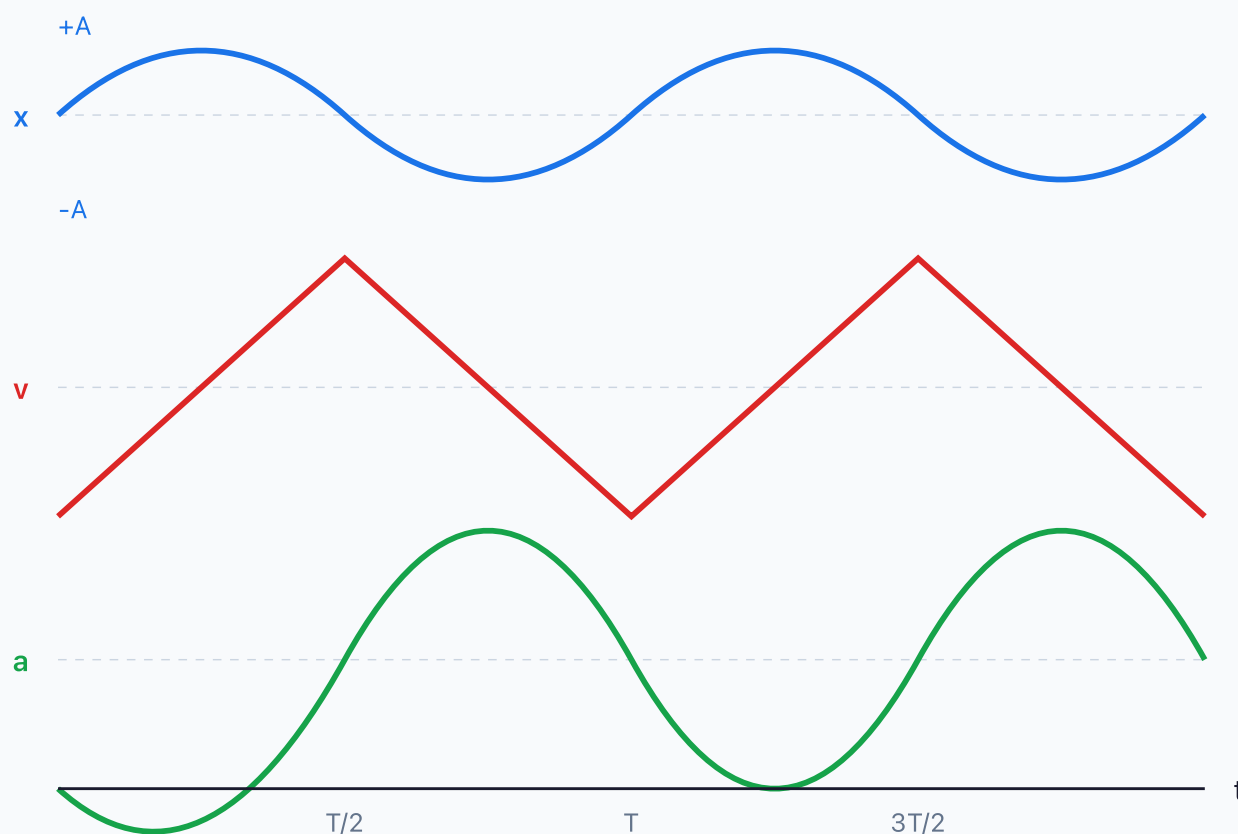
$$a = -\omega^2 x$$

The negative sign indicates the acceleration is always in the opposite direction to the displacement (i.e. it is a restoring acceleration, always pulling the object back towards the equilibrium position).

SHM Equations

Quantity	Equation	Notes
Displacement	$x = A \cos(\omega t)$ $x = A \sin(\omega t)$	Use cos if timing starts at max displacement; use sin if timing starts at equilibrium
Velocity	$v = -A\omega \sin(\omega t)$ $v = \pm \omega \sqrt{A^2 - x^2}$	The second form is very useful for finding v at a given x without knowing t
Maximum velocity	$v_{\max} = A\omega$	Occurs at $x = 0$ (the equilibrium position)
Maximum acceleration	$a_{\max} = A\omega^2$	Occurs at $x = \pm A$ (the extremes)
Angular frequency	$\omega = 2\pi f = 2\pi/T$	Always in rad s^{-1}

SHM: Displacement, Velocity, and Acceleration vs Time



Velocity leads displacement by $\pi/2$. Acceleration is in antiphase with displacement.

Phase Relationships

- Velocity leads displacement by $\pi/2$ radians (90 degrees, or a quarter of a period).
- Acceleration is in antiphase with displacement (π radians or 180 degrees out of phase).
- When $x = 0$, $v = \pm v_{\max}$ and $a = 0$.
- When $x = \pm A$, $v = 0$ and $a = \mp a_{\max}$.

Worked Example: A mass on a spring oscillates with SHM. The amplitude is 0.050 m and the period is 0.40 s. Calculate (a) the angular frequency, (b) the maximum velocity, (c) the maximum acceleration, (d) the velocity when the displacement is 0.030 m.

Solution:

(a) $\omega = 2\pi/T = 2\pi/0.40 = \mathbf{15.7 \text{ rad s}^{-1}}$

(b) $v_{\max} = A\omega = 0.050 \times 15.7 = \mathbf{0.79 \text{ m s}^{-1}}$

(c) $a_{\max} = A\omega^2 = 0.050 \times (15.7)^2 = 0.050 \times 247 = \mathbf{12.3 \text{ m s}^{-2}}$

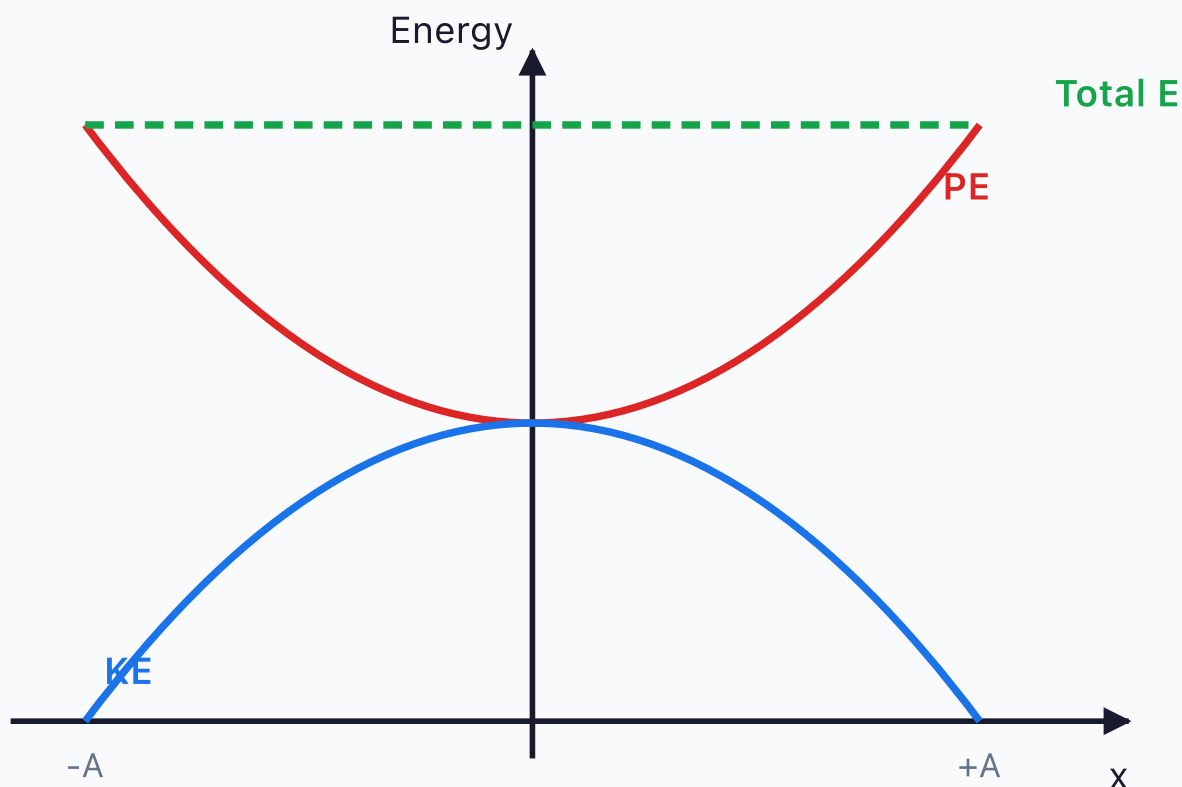
$$(d) v = \omega \sqrt{(A^2 - x^2)} = 15.7 \times \sqrt{(0.050^2 - 0.030^2)} = 15.7 \times \sqrt{(0.0025 - 0.0009)} = 15.7 \times \sqrt{(0.0016)} = 15.7 \times 0.040 = \mathbf{0.63 \text{ m s}^{-1}}$$

Energy in SHM

- At the equilibrium ($x = 0$): KE is maximum, PE is zero.
- At the extremes ($x = \pm A$): KE is zero, PE is maximum.
- Total energy = KE + PE = constant (in the absence of damping).
- $E_{\text{total}} = \frac{1}{2}m\omega^2 A^2$
- KE at displacement x : $E_k = \frac{1}{2}m\omega^2(A^2 - x^2)$
- PE at displacement x : $E_p = \frac{1}{2}m\omega^2 x^2$

The total energy is proportional to A^2 . Doubling the amplitude quadruples the total energy.

Energy vs Displacement in SHM



$$\text{Total energy} = \text{KE} + \text{PE} = \text{constant} = \frac{1}{2}m\omega^2 A^2$$

Time Periods of SHM Systems

Mass-spring system:

$$T = 2\pi\sqrt{m/k}$$

T depends on mass m and spring constant k . It does NOT depend on amplitude or on g .

Simple pendulum:

$$T = 2\pi\sqrt{L/g}$$

T depends on length L and gravitational field strength g . It does NOT depend on mass or amplitude (for small angles, typically < 10 degrees).

Exam Tip: In SHM, the period is independent of amplitude. This is a defining feature called **isochronous oscillation**. If the period changes with amplitude, the motion is not truly SHM. For the pendulum, the small-angle approximation $\sin\theta \approx \theta$ (in radians) is needed for the motion to be SHM.

Exam Tip: When verifying SHM from the defining equation, you need to show that $a \propto -x$. For a mass-spring: $F = -kx$, so $a = -(k/m)x$, confirming SHM with $\omega^2 = k/m$. For a pendulum: the restoring force along the arc is $F = -mg \sin\theta \approx -mg\theta = -mg(x/L)$, giving $a = -(g/L)x$, confirming SHM with $\omega^2 = g/L$.

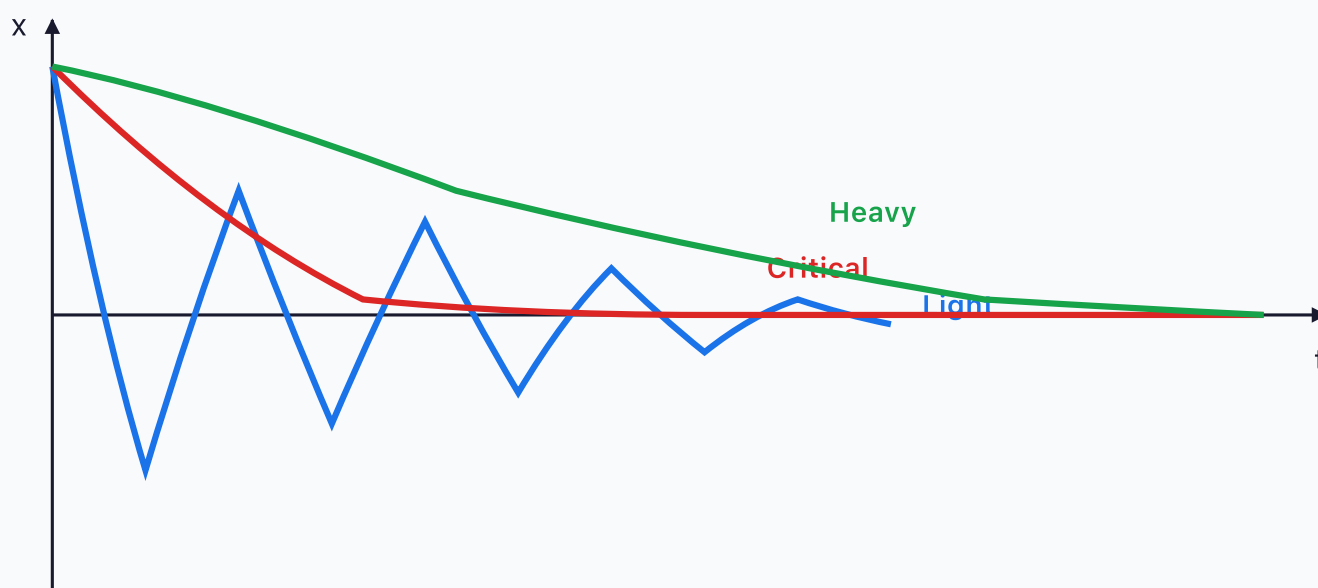
3. Damping

Damping: The reduction in amplitude of an oscillation over time due to energy being transferred to the surroundings (usually as thermal energy due to resistive or frictional forces). The frequency of a damped oscillation is slightly lower than the natural frequency.

Type of Damping	Description	Example
Light (under) damping	Amplitude decreases gradually over many oscillations. The system oscillates with slowly decreasing amplitude, with the decay envelope being exponential. The period is hardly affected.	Pendulum swinging in air; vibrating guitar string

Heavy (over) damping	The system returns to equilibrium very slowly without oscillating. The restoring force is overwhelmed by the damping force.	Door closer on a heavy fire door; shock absorber in thick oil
Critical damping	The system returns to equilibrium in the shortest possible time without oscillating past the equilibrium position. This is the boundary between under-damping and over-damping.	Car suspension (designed for near-critical damping); galvanometer needle

Types of Damping



Critical damping reaches equilibrium fastest without overshooting

Forced Oscillations and Resonance

Free oscillation: An oscillation where the system vibrates at its natural frequency with no external driving force (though damping may be present).

Forced oscillation: An oscillation caused by an external periodic driving force. The system oscillates at the driving frequency, not necessarily its natural frequency.

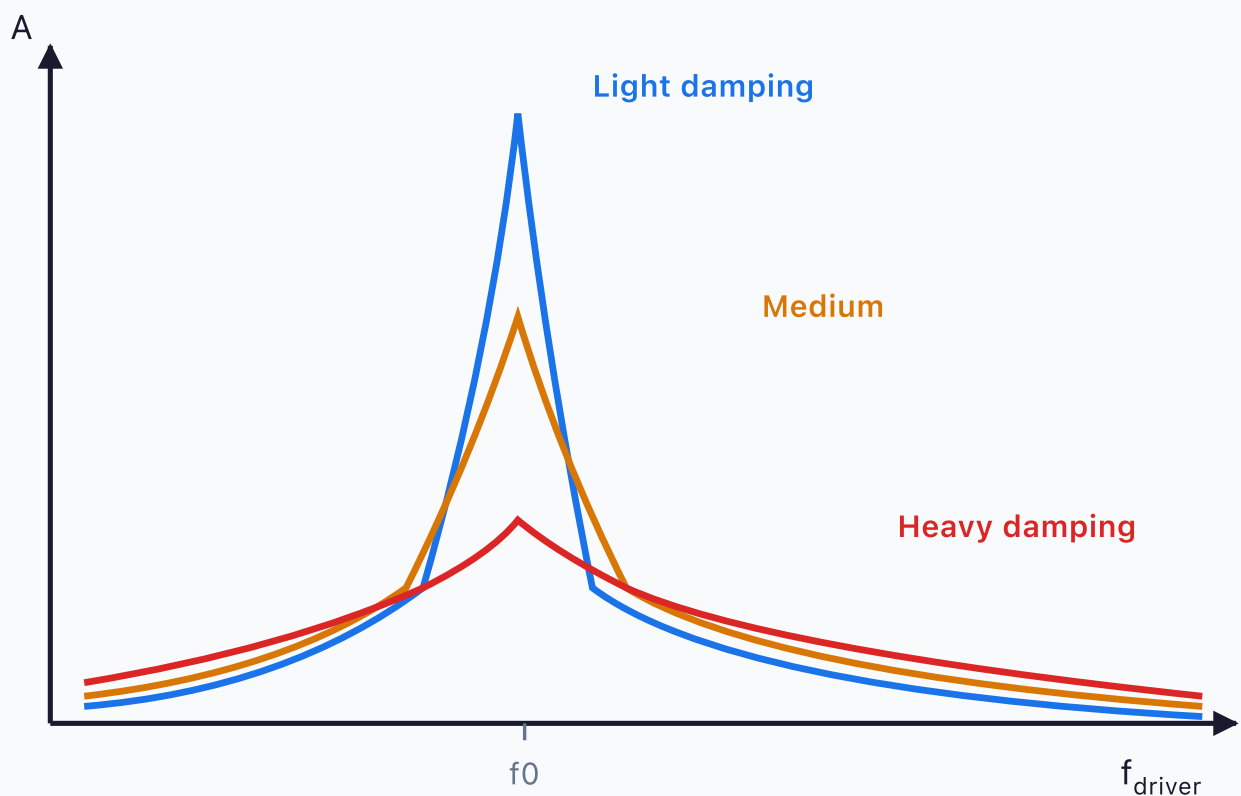
Natural frequency (f_0): The frequency at which a system oscillates freely when displaced and released (with no damping).

Resonance: Occurs when the driving frequency equals the natural frequency of the system ($f_{\text{driver}} = f_0$). At resonance, maximum energy is transferred from the driver to the oscillating system, resulting in maximum amplitude of oscillation.

Effects of Damping on Resonance

- Increased damping **reduces the maximum amplitude** at resonance.
- Increased damping **broadens the resonance peak** (the response curve becomes flatter and wider).
- Increased damping **shifts the resonance frequency slightly below f_0** .
- With very heavy damping, the resonance peak disappears entirely.

Resonance Curves (Amplitude vs Driving Frequency)



More damping: lower peak, broader curve, peak shifts slightly left

4. Thermal Physics

Internal Energy

Internal energy: The sum of the randomly distributed kinetic energies and potential energies of all the molecules in a system.

- The kinetic energy component is associated with the random translational, rotational, and vibrational motion of molecules
- The potential energy component is associated with the intermolecular forces between molecules
- For an ideal gas, there are no intermolecular forces, so internal energy is purely kinetic

The **First Law of Thermodynamics** states that a change in internal energy equals the heat added to the system minus the work done by the system: $\Delta U = Q - W$. Internal energy can change by:

- Doing work on or by the system (compressing or expanding a gas)
- Heating or cooling the system (energy transfer due to a temperature difference)

Temperature

Temperature is a measure of the average kinetic energy of the molecules. Two objects are in thermal equilibrium when they are at the same temperature (no net heat flow between them). The absolute (Kelvin) scale: $T(K) = T(^{\circ}C) + 273.15$. Absolute zero (0 K) is the lowest possible temperature, at which molecules have minimum (zero-point) energy.

Specific Heat Capacity

Specific heat capacity (c): The energy required to raise the temperature of 1 kg of a substance by 1 K (or 1 $^{\circ}C$). Units: $J\ kg^{-1}\ K^{-1}$.

$$Q = mc\Delta\theta$$

- Q = energy transferred (J)
- m = mass (kg)
- c = specific heat capacity ($J\ kg^{-1}\ K^{-1}$)
- $\Delta\theta$ = change in temperature (K or $^{\circ}C$ — both work since the intervals are equal)

Substance	Specific heat capacity ($J\ kg^{-1}\ K^{-1}$)
-----------	---

Water	4200
Aluminium	900
Copper	390
Iron/Steel	450
Air	~1000

Worked Example: A 2.0 kW kettle contains 1.5 kg of water at 20 °C. How long does it take to heat the water to 100 °C? Assume all energy goes into heating the water (no losses).

Solution:

$$Q = mc\Delta\theta = 1.5 \times 4200 \times (100 - 20) = 1.5 \times 4200 \times 80 = 504000 \text{ J}$$

$$P = Q/t, \text{ so } t = Q/P = 504000 / 2000 = \mathbf{252 \text{ s}}$$
 (about 4.2 minutes)

In practice it takes longer because energy is lost to the surroundings and to heating the kettle itself.

Specific Latent Heat

Specific latent heat (L): The energy required to change the state of 1 kg of a substance at constant temperature. Units: J kg^{-1} .

$$Q = mL$$

- **Specific latent heat of fusion (L_f):** for solid $\leftrightarrow$ liquid transitions
- **Specific latent heat of vaporisation (L_v):** for liquid $\leftrightarrow$ gas transitions

During a change of state, temperature remains constant because the energy supplied is used to break (or form) intermolecular bonds, changing potential energy without changing kinetic energy. L_v is always much larger than L_f because the molecules must be completely separated during vaporisation.

Substance	L_f (kJ kg^{-1})	L_v (kJ kg^{-1})
Water	334	2260

Aluminium	397	10900
-----------	-----	-------

Exam Tip: In a heating curve (temperature vs time at constant power), the flat (horizontal) sections represent changes of state where latent heat is being supplied. The gradient of the sloped sections depends on specific heat capacity. A lower gradient = higher specific heat capacity (for the same power input and mass). The length of a flat section depends on the latent heat — a longer plateau means a higher specific latent heat.

5. Gas Laws and the Ideal Gas

The Three Gas Laws

Law	Statement	Equation	Conditions
Boyle's Law	Pressure is inversely proportional to volume	$pV = \text{constant}$	Constant temperature (isothermal) and fixed mass of gas
Charles's Law	Volume is directly proportional to absolute temperature	$V/T = \text{constant}$	Constant pressure (isobaric) and fixed mass
Pressure Law (Gay-Lussac's)	Pressure is directly proportional to absolute temperature	$p/T = \text{constant}$	Constant volume (isochoric) and fixed mass

These combine into the combined gas law:

$$p_1 V_1 / T_1 = p_2 V_2 / T_2$$

Ideal Gas Equation

$$pV = nRT = NkT$$

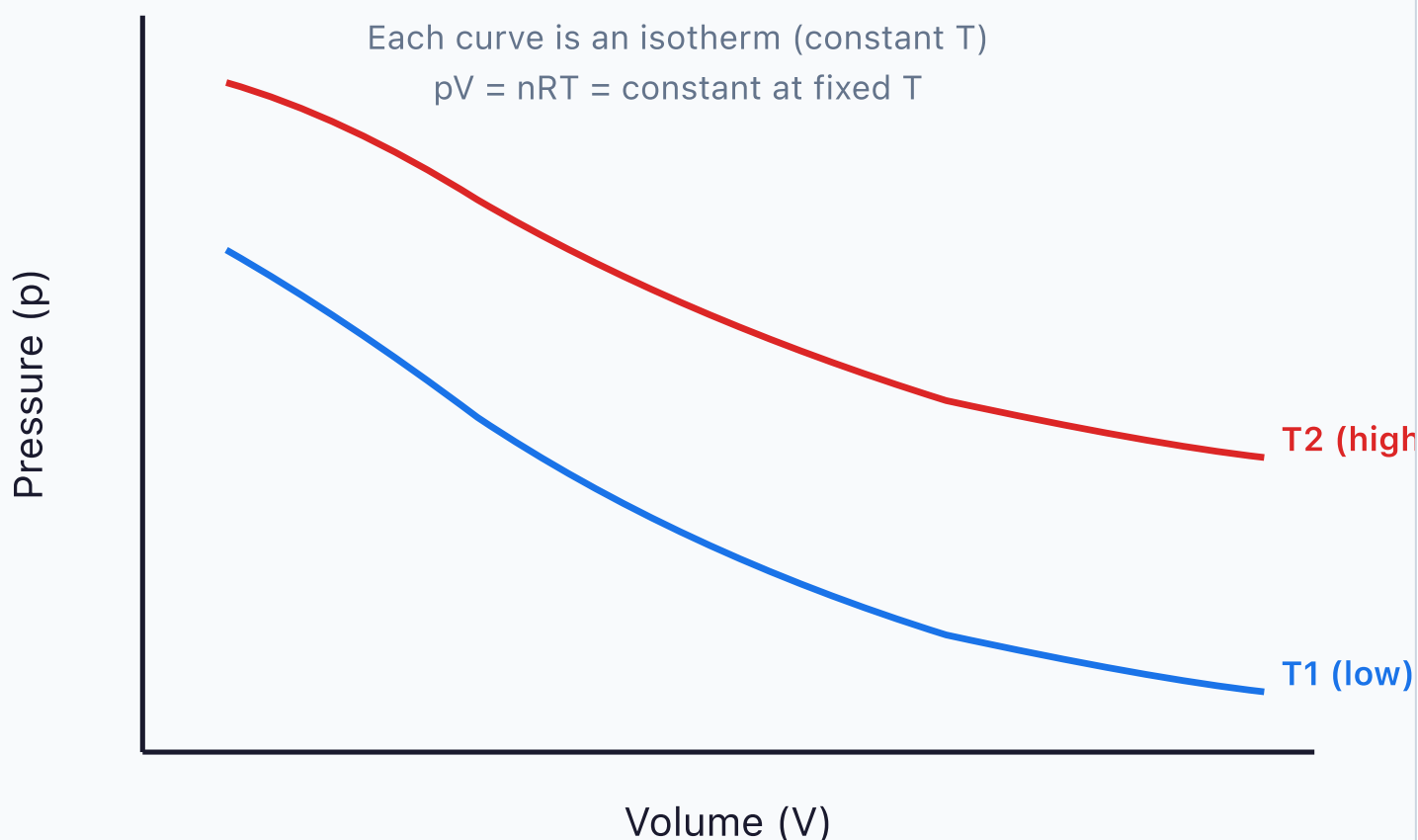
- p = pressure (Pa)
- V = volume (m^3)

- n = number of moles (mol)
- R = molar gas constant = $8.31 \text{ J mol}^{-1} \text{ K}^{-1}$
- N = number of molecules
- k = Boltzmann constant = $1.38 \times 10^{-23} \text{ J K}^{-1}$
- T = absolute temperature (K)

Relationships: $N = nN_A$ and $k = R/N_A$, where $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ (Avogadro constant).

Number of moles: $n = m/M$ (mass / molar mass) = N/N_A .

Pressure-Volume Diagram (Boyle's Law Isotherms)



Worked Example: A sealed container holds 0.50 mol of an ideal gas at 300 K and $1.0 \times 10^5 \text{ Pa}$.
(a) Calculate the volume of the gas. (b) The gas is heated to 450 K at constant volume. Calculate the new pressure.

Solution:

(a) $pV = nRT$, so $V = nRT/p = (0.50 \times 8.31 \times 300) / (1.0 \times 10^5) = 1246.5 / 100000 = \mathbf{0.0125 \text{ m}^3}$
(12.5 litres)

(b) At constant volume and fixed n : $p_1/T_1 = p_2/T_2$

$$p_2 = p_1 \times T_2/T_1 = 1.0 \times 10^5 \times 450/300 = \mathbf{1.5 \times 10^5 \text{ Pa}}$$

Exam Tip: Always convert temperature to kelvin ($K = ^\circ\text{C} + 273$) and volume to m^3 (1 litre = 10^{-3} m^3 , $1 \text{ cm}^3 = 10^{-6} \text{ m}^3$) before using the ideal gas equation. Pressure must be in Pa (1 atm = $1.013 \times 10^5 \text{ Pa}$).

6. Kinetic Theory of Gases

Assumptions of the Kinetic Theory (Ideal Gas Model)

1. The gas contains a **very large number** of molecules in **random** motion.
2. The **volume of the molecules is negligible** compared to the volume of the container.
3. Collisions between molecules and between molecules and the walls are **perfectly elastic** (kinetic energy is conserved).
4. The **time of a collision is negligible** compared to the time between collisions.
5. There are **no intermolecular forces** between molecules except during collisions (no attractive or repulsive forces at a distance).

Kinetic Theory Equation

$$pV = \frac{1}{3}Nmc^2$$

where m is the mass of one molecule, N is the number of molecules, and $\overline{c^2}$ is the mean square speed of the molecules.

Root Mean Square (rms) Speed

$$c_{\text{rms}} = \sqrt{\overline{c^2}}$$

The rms speed is NOT the same as the mean speed. It gives more weight to faster molecules. The rms speed is always slightly larger than the mean speed for a Maxwell-Boltzmann distribution.

Molecular Kinetic Energy and Temperature

Combining the kinetic theory equation with the ideal gas equation ($pV = NkT = \frac{1}{3}Nm\overline{c^2}$) gives:

$$\frac{1}{2}m\overline{c^2} = \frac{3}{2}kT$$

This is a profoundly important result: the average translational kinetic energy of a gas molecule is directly proportional to the absolute temperature. This connects the microscopic (molecular KE) to the macroscopic (temperature).

For one mole of gas, the total translational KE is:

$$E_{\text{total}} = \frac{3}{2}nRT = \frac{3}{2}NkT$$

Worked Example: Calculate (a) the average KE of a nitrogen molecule at 300 K, (b) the rms speed of nitrogen molecules at 300 K. ($m_{\text{N}_2} = 28 \times 1.66 \times 10^{-27} \text{ kg} = 4.65 \times 10^{-26} \text{ kg}$)

Solution:

$$(a) \frac{1}{2}m\overline{c^2} = \frac{3}{2}kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300 = \mathbf{6.21 \times 10^{-21} \text{ J}}$$

$$(b) \overline{c^2} = 3kT/m = (3 \times 1.38 \times 10^{-23} \times 300) / (4.65 \times 10^{-26}) = 1.242 \times 10^{-20} / 4.65 \times 10^{-26} = 2.67 \times 10^5 \text{ m}^2 \text{ s}^{-2}$$

$$c_{\text{rms}} = \sqrt{(2.67 \times 10^5)} = \mathbf{517 \text{ m s}^{-1}}$$

This is faster than the speed of sound in air ($\sim 340 \text{ m s}^{-1}$), which makes sense since sound propagates through molecular collisions.

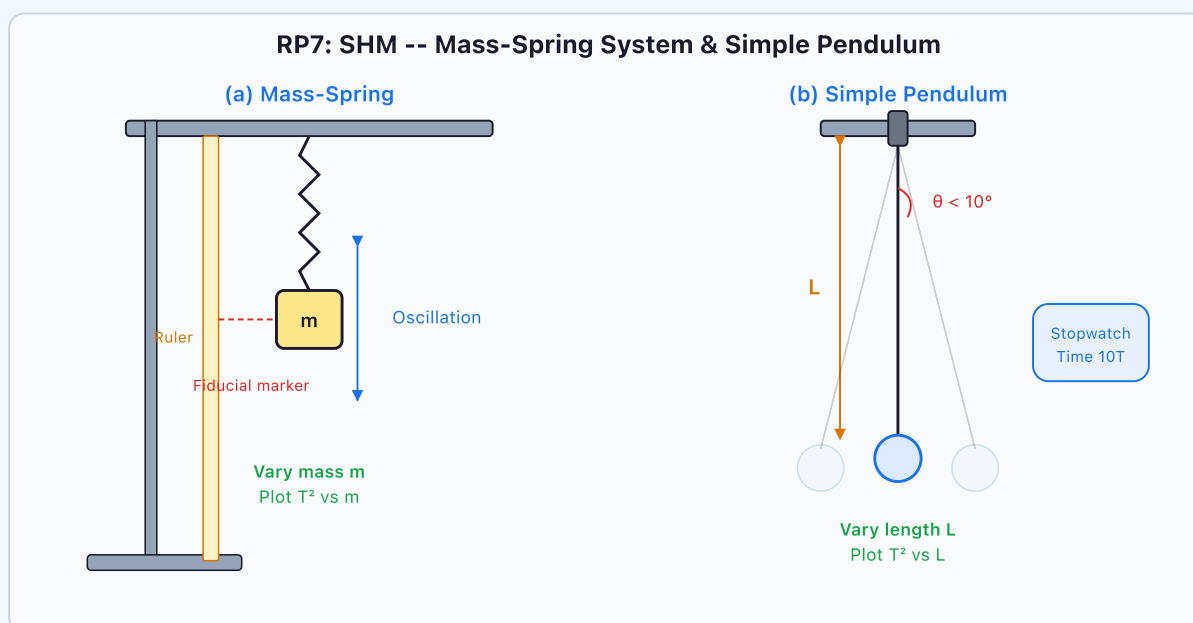
Exam Tip: The Boltzmann constant k links microscopic energy to temperature. If a question asks about energy *per molecule*, use $\frac{3}{2}kT$. If it asks about energy *per mole*, use $\frac{3}{2}RT$. Remember that temperature must always be in kelvin. Doubling the absolute temperature doubles the average KE, but the rms speed only increases by a factor of $\sqrt{2}$.

Summary of Key Equations

Quantity	Equation
Circular motion	$a = v^2/r = \omega^2 r; \quad F = mv^2/r = m\omega^2 r$

SHM defining equation	$a = -\omega^2 x$
SHM displacement	$x = A \cos(\omega t)$ or $x = A \sin(\omega t)$
SHM velocity	$v = \pm \omega \sqrt{A^2 - x^2}$; $v_{\max} = A\omega$
Mass-spring period	$T = 2\pi\sqrt{m/k}$
Simple pendulum period	$T = 2\pi\sqrt{L/g}$
Specific heat capacity	$Q = mc\Delta\theta$
Specific latent heat	$Q = mL$
Ideal gas equation	$pV = nRT = NkT$
Kinetic theory	$pV = \frac{1}{3}Nm\overline{c^2}$
Molecular KE	$\frac{1}{2}m\overline{c^2} = \frac{3}{2}kT$

Required Practical 7: Investigation into SHM (Mass-Spring & Pendulum)



Apparatus

(a) Mass-Spring: Helical spring, slotted masses and hanger, retort stand with boss and clamp, metre ruler with fiducial marker, stopwatch (0.01 s resolution).

(b) Simple Pendulum: String (at least 1 m, light and inextensible), pendulum bob (small, dense sphere), clamp with boss and retort stand, metre ruler or tape measure, protractor (to check angle $< 10^\circ$), stopwatch.

Method

(a) Mass-Spring:

1. Hang the spring from the clamp. Attach a known mass m and allow it to reach equilibrium. Mark the equilibrium position with a fiducial marker on the ruler.
2. Pull the mass down a small distance (< 5 cm) and release. Time 10 complete oscillations with the stopwatch. Divide by 10 to get T .
3. Repeat 3 times and calculate a mean value of T at each mass.
4. Change the mass in steps (e.g. 50 g increments from 100 g to 500 g). Record T for each mass.
5. Plot T^2 (y-axis) vs m (x-axis). The gradient $= 4\pi^2/k$, so $k = 4\pi^2/\text{gradient}$.

(b) Simple Pendulum:

1. Set up the pendulum with a measured length L (from the pivot point to the centre of the bob).
2. Displace the bob by a small angle ($< 10^\circ$, checked with a protractor) and release. Time 10 complete oscillations. Divide by 10 to get T .
3. Repeat 3 times and calculate a mean T .
4. Change L in steps (e.g. 0.10 m from 0.20 m to 1.00 m). Record T at each length.
5. Plot T^2 (y-axis) vs L (x-axis). The gradient $= 4\pi^2/g$, so $g = 4\pi^2/\text{gradient}$.

Safety

- Ensure the retort stand is clamped securely to the bench with a G-clamp to prevent toppling.
- Place a cushion beneath hanging masses in case the spring breaks.
- Keep small amplitudes to ensure SHM conditions are met.
- Stand clear of the swinging pendulum bob.

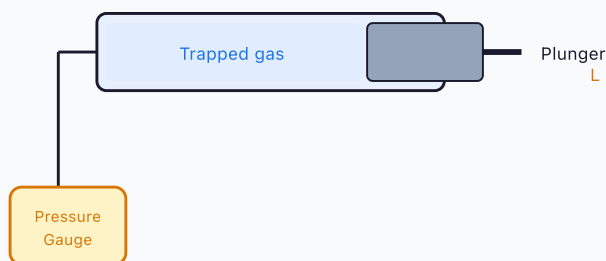
Sources of Error & Improvements

Source of Error	Type	Improvement
Human reaction time (~0.2 s)	Random	Time 10+ oscillations and divide; use a light gate or motion sensor
Amplitude too large (not truly SHM)	Systematic	Use small amplitudes ($< 10^\circ$ for pendulum, small extension for spring)
Pendulum length measured to wrong point on bob	Systematic	Measure to the centre of the bob; use a small, dense bob
Spring oscillates non-vertically (wobbling)	Random	Guide the mass vertically; use a card or guide tube
Counting error (miscounting oscillations)	Random	Use a fiducial marker; start counting from zero
Effective mass of the spring contributes to period	Systematic	Use a light spring; the y-intercept of T^2 vs m accounts for the spring's effective mass ($\frac{1}{3}$ of spring mass)

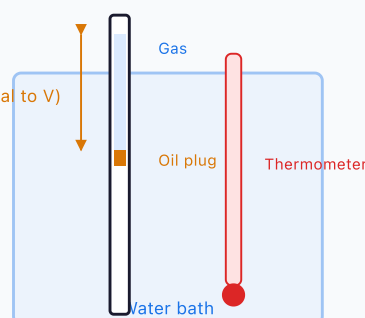
Exam Tip: Always time at least 10 oscillations and divide to reduce the percentage uncertainty from reaction time. Start counting from **zero** (not one) as the mass passes the fiducial marker. Plotting T^2 (not T) gives a straight-line graph, making it easier to determine the gradient and hence k or g .

Required Practical 8: Boyle's Law and Charles's Law

RP8: Boyle's Law and Charles's Law

(a) Boyle's Law: P vs $1/V$ 

Push plunger: vary V
Record P at each V
Plot P vs $1/V$

(b) Charles's Law: V vs T 

Bunsen / hot plate
Plot V (or L) vs T (in K)

Apparatus

(a) Boyle's Law: Gas syringe (sealed) with oil plug, Bourdon pressure gauge or digital pressure sensor, clamp and stand.

(b) Charles's Law: Sealed capillary tube with trapped air column (sealed by oil drop), water bath (beaker of water), thermometer ($-10\text{ }^{\circ}\text{C}$ to $110\text{ }^{\circ}\text{C}$) or temperature sensor, Bunsen burner or electric hot plate, ruler, stirrer.

Method

(a) Boyle's Law (constant temperature):

1. Record the initial volume V of trapped gas from the syringe scale and the pressure P from the gauge.
2. Slowly push the plunger in to reduce the volume. Wait for the temperature to equalise (so the process is isothermal), then record V and P .
3. Repeat for several different volumes, both increasing and decreasing the volume.
4. Plot P (y-axis) vs $1/V$ (x-axis). A straight line through the origin confirms Boyle's Law.

(b) Charles's Law (constant pressure):

1. Place the capillary tube in the water bath. Record the initial temperature T and length L of the gas column (L is proportional to V since the tube has uniform cross-section).
2. Heat the water slowly, stirring continuously. At regular intervals (e.g. every $10\text{ }^{\circ}\text{C}$), record T and L .

3. Continue until near boiling. Allow to cool and take readings on the way down to check for consistency.
4. Convert temperatures to kelvin: $T(K) = T(^{\circ}C) + 273.15$.
5. Plot V (or L) against T (in kelvin). A straight line that, when extrapolated, passes through the origin confirms V is proportional to T .

Safety

- Take care with hot water; use tongs or clamps to handle the capillary tube.
- If using concentrated sulfuric acid as a plug, handle with extreme care (gloves and goggles).
- Do not heat above $100^{\circ}C$; the tube may crack or the plug may be expelled.
- Ensure the gas syringe is clamped securely.

Sources of Error & Improvements

Source of Error	Type	Improvement
Temperature not uniform in water bath	Systematic	Stir continuously; allow time to equilibrate
Gas leaks past the oil plug or syringe seal	Systematic	Use a good seal; check for leaks by repeating readings
Friction in the syringe affects pressure readings	Systematic	Tap the syringe gently; use a low-friction syringe
Gas may not behave ideally at high pressures	Systematic	Keep within moderate pressure and temperature ranges
Reading the gas column length is imprecise	Random	Use a magnifying glass; read from the same angle

Exam Tip: For Boyle's law, plot P vs $1/V$ (not P vs V , which gives a curve). For Charles's law, always convert to kelvin before plotting. Extrapolating the V vs T graph to $V = 0$

should give a temperature intercept near $-273\text{ }^{\circ}\text{C}$, providing an experimental estimate of absolute zero.